WHITE PAPER

Steady-State and Time-Varying Queueing Analysis of a Theme Park Transit System:

An M/M/c and Pointwise Stationary Approximation Approach to the Disney World Monorail

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github.com/hamkhadafi/disney-monorail-queue-analysis

Abstract

Theme park transit systems present a tractable real-world setting for applied queueing analysis: high passenger volume, multi-unit capacity, and publicly observable wait-time data. This paper models the Disney World Monorail boarding queue as a Continuous-Time Markov Chain under the M/M/c framework, using 30,108 fifteen-minute interval records spanning January 2018 to March 2020. Maximum-likelihood parameter estimation yields an arrival rate of 9.11 guests/min and a per-unit service rate of 0.86 guests/min, with c=11 boarding units operating at 96.2% utilization—a near-critical steady state with mean queue length of 21.9 guests and mean wait of 144.6 seconds. A chi-squared goodness-of-fit test rejects the homogeneous Poisson arrival assumption (variance-to-mean ratio = 7.76), motivating a Pointwise Stationary Approximation (PSA) extension that recomputes queueing metrics using hour-specific arrival rates. This reveals that the system exceeds capacity ($\rho > 1$) during the 09:00–10:00 morning peak, a finding invisible to the global M/M/c baseline. Sensitivity analysis across $c \in \{9,...,15\}$ shows that increasing capacity to 12 units reduces baseline wait time by 80.1%, while 13 units restores stability during peak hours. These results demonstrate how combining steady-state and time-varying queueing models yields capacity planning insight that neither approach provides alone.

Keywords—*queueing theory, M/M/c, Continuous-Time Markov Chain, Erlang-C, pointwise stationary approximation, capacity planning*

1. INTRODUCTION

Queueing theory provides a rigorous mathematical framework for analyzing systems where demand for service is stochastic and capacity is finite. While much of the queueing literature is motivated by telecommunications, call centers, and healthcare systems, theme park transit systems offer an underexplored but analytically rich setting: high-frequency, high-volume demand; discrete, countable service units; and the rare availability of granular, publicly observable operational data.

The Disney World Monorail is a fixed-route transit system connecting the resort's parks and hotels. Passenger demand fluctuates substantially throughout the operating day, while the number of active monorail units is comparatively stable. This combination—variable demand against quasi-fixed capacity—is the canonical setting for multi-server queueing analysis.

This paper addresses three objectives:

- Model the Monorail boarding queue as an M/M/c system, estimate its parameters via maximum likelihood, and derive steady-state performance metrics using Continuous-Time Markov Chain (CTMC) theory.
- Test the validity of core M/M/c assumptions against observed data, and extend the model to account for time-varying demand where the stationary assumption fails.
- Translate model output into actionable capacity planning recommendations, validated against observed wait-time data.

The analysis is intended to demonstrate not only the mechanics of CTMC-based queueing models, but the applied judgment required when real data violates textbook assumptions—a gap between theoretical queueing models and their practical deployment that is rarely addressed explicitly in introductory treatments.

2. DATASET

The dataset is sourced from the “Disney World Ride Wait Times” collection on Kaggle [1], comprising 1,317,071 fifteen-minute interval records across 37 attractions from January 1, 2018 to March 13, 2020. This analysis filters to the Monorail attraction exclusively, yielding 30,108 records after removing intervals with non-positive duration or zero guests carried.

Table 1. Key Dataset Fields (Monorail Subset)

Field	Description
DEB_TIME / FIN_TIME	Interval start and end timestamps (nominal 15-minute intervals)
GUEST_CARRIED	Number of guests served within the interval — used to estimate λ
NB_UNITS	Number of monorail units in operation — used to estimate c
UP_TIME / OPEN_TIME	Operational uptime and scheduled open time within the interval (minutes) — used to construct an availability proxy for ρ
WAIT_TIME_MAX	Posted maximum wait time (minutes) — used for external validation only

The number of operating units, c, is estimated as the mode of NB_UNITS across all retained records, yielding c = 11. This is consistent with public reporting that the Monorail fleet operates with a subset of its total rolling stock active at any given time, with the remainder held in reserve or maintenance rotation.

3. METHODOLOGY

3.1 M/M/c Framework

The boarding queue is modeled as an M/M/c queue: Poisson arrivals at rate λ, c parallel exponential servers each with service rate μ, and a single FCFS queue feeding all servers. Under this model, the number of guests in the system, N(t), is a birth-death Continuous-Time Markov Chain with state-dependent transition rates [2]:

- Birth rate (arrival): λ for all states $n \geq 0$
- Death rate (departure): $\min(n, c) \cdot \mu$ at state n

The system is stable if and only if the traffic intensity $\rho = \lambda / (c\mu) < 1$, which is required for a proper steady-state distribution to exist.

3.2 Parameter Estimation

Parameters are estimated via maximum likelihood on the filtered interval data:

$\hat{\lambda}$ (arrival rate): computed as GUEST_CARRIED / interval_min for each record, then averaged across all intervals, yielding $\hat{\lambda}$ = 9.1056 guests/min.

c (number of servers): the mode of NB_UNITS, yielding c = 11.

$\hat{\rho}$ (utilization proxy): computed per-interval as UP_TIME / OPEN_TIME, clipped to [0.01, 0.99] to avoid degenerate values.

$\hat{\mu}$ (service rate per server): derived algebraically as $\hat{\lambda}$ / (c · $\hat{\rho}$), yielding $\hat{\mu}$ = 0.8602 guests/min/server.

A methodological caveat is necessary here. The OPEN_TIME and UP_TIME fields in the source dataset are not formally documented; no data dictionary accompanies the Kaggle release. Based on standard naming conventions in theme-park operational systems, OPEN_TIME most plausibly represents the duration within an interval that the ride was scheduled to operate, while UP_TIME represents the duration the ride was actually running (i.e., not in a breakdown or maintenance stop). Under this interpretation, $\hat{\rho}$ = UP_TIME/OPEN_TIME measures operational availability—the fraction of time the ride was not down—rather than a directly observed server busy fraction in the queueing-theoretic sense. Consequently, $\hat{\mu}$ derived from this proxy should be understood as an estimate consistent with an M/M/c fit to aggregate throughput, not a measurement of literal per-unit service time. This distinction does not invalidate the modeling approach—parameter estimation via an algebraically consistent proxy is standard practice when direct service-time observations are unavailable—but it should temper any claim that $\hat{\mu}$ represents a directly measured quantity.

The resulting offered load is r = λ/μ = 10.585, and per-server utilization is ρ = λ/(cμ) = 0.9623 (96.23%). Since ρ < 1, the system satisfies the stability condition.

3.3 Steady-State Formulas

Given λ, μ, and c, the steady-state probability of an empty system, P_0, and the Erlang-C probability of delay, C(c,r), follow the standard closed-form expressions for M/M/c queues [2]:

> $P_0 = [\ \Sigma_{n=0}$^(c−1) $(r^n/n!) + (r^c/c!)\cdot(1/(1-\rho))\ ]^{-1}$, $C(c,r) = (r^c\ P_0)$ / $(c!\ (1-\rho))$

From these, the mean queue length, waiting time, and total system metrics follow via the Erlang-C relations and Little’s Law:

- Lq = C(c,r) · ρ / (1−ρ) (mean number waiting)
- Wq = Lq / λ (mean waiting time, by Little’s Law)
- L = Lq + r, W = Wq + 1/μ (mean number and time in system)

The full steady-state distribution P(n) is computed recursively from the birth-death balance equations and used to characterize the probability mass beyond the c-server boundary.

3.4 Goodness-of-Fit and Motivation for a Time-Varying Extension

The M/M/c model assumes Poisson arrivals, which implies that GUEST_CARRIED counts should be approximately Poisson-distributed with variance equal to the mean. This assumption is tested via a chi-squared goodness-of-fit test against a fitted Poisson(λ=137) distribution on interval-level guest counts.

The test strongly rejects the Poisson hypothesis ($\chi^2 = 1{,}336{,}114$, df = 82, $p \approx 0$), with a maximum standardized residual of 897.6—far beyond what large-sample artifacts would produce. The variance-to-mean ratio (VMR) of 7.76 confirms substantial overdispersion relative to the Poisson benchmark (VMR ≈ 1).

This overdispersion is not random noise; it reflects systematic, predictable structure. Mean guests carried per interval varies from a peak of approximately 158 at 09:00 to a trough of approximately 108 at 21:00—a pattern of non-stationary demand that a single global λ cannot capture. Rather than treating this as a fatal flaw, this analysis retains the global M/M/c model as a local-stationarity approximation—standard practice in applied queueing analysis [2]—and extends it in Section 4.3 with an hour-specific Pointwise Stationary Approximation that directly addresses the non-stationarity the goodness-of-fit test reveals.

4. RESULTS

4.1 Baseline M/M/c Steady-State Metrics

Table 2 reports the steady-state performance metrics under the global (24-hour-average) M/M/c model.

Table 2. Baseline M/M/c Steady-State Metrics (c=11)

Metric	Value
$\hat{\lambda}$ (arrival rate)	9.1056 guests/min
$\hat{\mu}$ (service rate per unit)	0.8602 guests/min
Offered load ($r = \lambda/\mu$)	10.585
Utilization (ρ)	96.23%
P_0 (probability system empty)	0.0007%
Erlang-C, C(c,r) (prob. of delay)	86.02%
Lq (mean number waiting)	21.94 guests
L (mean number in system)	32.53 guests
Wq (mean wait time)	2.41 min (144.6 sec)
W (mean time in system)	3.57 min (214.3 sec)

At ρ = 96.23%, the system operates with virtually no slack capacity. The Erlang-C probability of 86.02% indicates that the large majority of arriving guests encounter a queue rather than immediate boarding—a hallmark of a system operating near saturation.

4.2 Steady-State Distribution P(n)

The full distribution $P(n)$ is heavy-tailed, a direct consequence of high utilization. The first 15 states capture only 26.2% of total probability mass; the distribution must be extended to 187 states to capture 99.9% of mass. The mode occurs at $n=10$ ($P=0.0337$), just below the server count. Critically, $P(n \le c=11) = 17.23\%$, meaning the system has an idle or non-queuing server available only 17.23% of the time—the remaining 82.77% of the time, at least one guest is waiting. This distributional result is consistent with, and numerically explains, the high Erlang-C delay probability reported above.

4.3 Time-Varying Extension: Pointwise Stationary Approximation

The goodness-of-fit results in Section 3.4 establish that a single global λ obscures substantial within-day variation. To address this, the analysis applies the Pointwise Stationary Approximation (PSA) of Green and Kolesar [3]: rather than assuming one stationary regime for the full operating day, PSA computes the steady-state M/M/c performance measures separately for each hour, using that hour's empirical arrival rate λ_h in place of the global $\hat{\lambda}$. This treats the system as locally stationary within each hour while allowing the equilibrium itself to shift across hours—directly modeling the non-stationarity that the chi-squared test identified, rather than averaging over it.

Applying hourly λ_h to the same $c=11$, $\hat{\mu} = 0.8602$ baseline reveals a critical finding invisible to the global model: at 09:00 and 10:00, the hourly arrival rate produces $\rho_h > 1$—the system is formally unstable during these hours, meaning the queue grows without bound for as long as the elevated arrival rate persists. The global M/M/c model, by averaging this peak against quieter hours, reports a stable system ($\rho=96.23\%$) that masks a genuine capacity shortfall during the morning peak.

This result reframes the relationship between Sections 4.1 and 4.3: the global M/M/c model provides a long-run average baseline, while the PSA extension provides the operationally relevant picture for staffing and capacity decisions, particularly during peak hours.

4.4 Model Validation

The model's theoretical W_q (2.41 minutes) is compared against WAIT_TIME_MAX, the posted maximum wait time in the dataset (mean = 61.47 min, median = 60 min). The substantial gap between these figures is expected rather than indicative of model failure: WAIT_TIME_MAX is a publicly displayed, strategically rounded communication to guests—commonly set conservatively above expected wait to manage expectations—not a measurement of actual per-guest queue time. This interpretation is supported by the near-zero correlation ($r = 0.02$) between the estimated utilization $\hat{\rho}$ and WAIT_TIME_MAX: if posted times tracked instantaneous queue state, a meaningfully positive correlation would be expected. Their near-independence indicates WAIT_TIME_MAX is set through an operational or marketing process largely decoupled from the real-time queueing dynamics this model estimates.

Hourly validation cross-checks the PSA finding in Section 4.3: at 09:00 and 10:00, computing ρ_h using hourly λ confirms $\rho_h > 1$, independently corroborating the morning instability identified through the PSA extension.

5. DISCUSSION

5.1 Capacity Planning

Table 3 reports sensitivity analysis across $c \in \{9,...,15\}$, holding $\hat{\lambda}$ and $\hat{\mu}$ fixed at their baseline estimates.

Table 3. Capacity Sensitivity Analysis

c	ρ	Wq (sec)	Wq Reduction
9	*117.6%*	*UNSTABLE*	*—*
10	*105.9%*	*UNSTABLE*	*—*
11 (baseline)	**96.2%**	**144.6**	**—**
12	88.2%	28.8	−80.1%
13	81.4%	11.1	−92.3%
14	75.6%	5.0	−96.6%
15	70.6%	2.4	−98.4%

The marginal returns to added capacity are steep and front-loaded: moving from c=11 to c=12 cuts mean wait time by 80.1%, while each subsequent unit yields diminishing absolute gains. Combined with the PSA finding in Section 4.3, this supports a two-tier recommendation: c=12 as a standing baseline capacity (reducing routine wait substantially while remaining operationally efficient), and c=13 specifically during the 09:00–11:00 morning peak to restore ρ below the stability threshold. A static c=15 deployment would virtually eliminate queueing ($W_q \approx 2.4$ sec) but at a utilization of only 70.6%—likely an inefficient allocation of capital assets (monorail trains) for the majority of off-peak hours.

5.2 Interpreting "Servers" in a Batch-Service System

A conceptual clarification is warranted regarding the physical interpretation of c and μ in this model. The Monorail does not serve guests one at a time per unit; each train departs in batches, carrying hundreds of guests per dispatch. The estimated $\hat{\mu} = 0.8602$ guests/min/server (approximately 70 seconds per guest per server) should therefore not be read as a literal single-passenger boarding time. Rather, the M/M/c decomposition here is a mathematical convention: $c \times \hat{\mu} \approx 9.46$ guests/min represents the system's aggregate throughput capacity, and its allocation across c abstract "servers" is a modeling device that maps a batch-dispatch process onto the standard multi-server queueing framework. Each "server" is best understood as an abstracted unit of boarding capacity contributing proportionally to system throughput, not as an independent agent processing one passenger at a time. This abstraction is standard practice when applying classical queueing models to batch-service transit systems, and does not undermine the validity of the aggregate performance metrics (W_q, L_q, ρ) derived from it—these depend only on the aggregate rate cμ, not on the literal mechanics of individual service completions.

5.3 Limitations

- The UP_TIME/OPEN_TIME-derived $\hat{\rho}$ is an operational availability proxy, not a directly observed server busy fraction; $\hat{\mu}$ inherits this approximation (Section 3.2).
- The Poisson arrival assumption is rejected at the aggregate level (VMR=7.76); the PSA extension mitigates but does not fully resolve this, as within-hour arrivals may still exhibit residual non-stationarity or correlation not captured by an hourly-constant rate.
- Aggregated 15-minute interval data precludes direct testing of the exponential service-time assumption, which is assumed rather than verified.
- The infinite-queue and homogeneous-server assumptions of the standard M/M/c model may not hold exactly in practice (e.g., physical queue space constraints, heterogeneous train capacities).
- WAIT_TIME_MAX validation is necessarily indirect, given the absence of individual-guest wait-time records in the source data.

6. CONCLUSION AND FUTURE WORK

This paper models the Disney World Monorail boarding queue using a CTMC-based M/M/c framework, deriving steady-state metrics from first principles and validating model assumptions against 30,108 observed intervals. The baseline model identifies a system operating at 96.23% utilization with a mean wait of 144.6 seconds. Rejection of the Poisson arrival assumption (VMR=7.76) motivates a Pointwise Stationary Approximation extension using hourly arrival rates, which reveals that the system is formally unstable ($\rho > 1$) during the 09:00–10:00 morning peak—a finding the global model cannot surface. Sensitivity analysis supports a capacity recommendation of c=12 as standing baseline and c=13 during morning peak hours, reducing baseline wait time by over 80% while restoring stability where the global model would otherwise mask a genuine shortfall.

Future work could extend this analysis in three directions: (1) acquiring or estimating individual-guest interarrival and service-time data to directly test the exponential assumptions underlying the CTMC formulation; (2) applying a full Mt/M/c(t) framework with continuously time-varying arrival rates, rather than the hourly-discretized PSA used here, to capture finer-grained demand transitions; and (3) extending the model to a network of queues representing the full Disney World transit system (monorail, buses, boats) to study cross-system capacity allocation under shared guest demand.

REFERENCES

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